SEAN WILKEN

Senior Software Engineer · Technical Leader

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SUMMARY

Senior Software Engineer with 10+ years of experience building production systems across healthcare, e-commerce, and AI-driven platforms. Experienced in delivering full-stack applications, data pipelines, and distributed systems from design through deployment. Strong background in F#, React, TypeScript and C#, with deep experience in regulated healthcare workflows, API design, and scalable web applications.

EXPERIENCE

Technical Co-Founder & Lead Engineer *January 2021 – Present*

Antidote AI *Orange, CA*

- Led back-end and full-stack development for an AI-powered healthcare platform deployed across multi-state medical facilities.
- Built diagnostic automation pipelines that processed 16M+ patient records, while maintaining strict HIPAA compliance.
- Designed and shipped secure data pipelines, cutting coordination overhead by 50% and improving clinical team throughput with up-to-date information.
- Built telehealth and chat workflows and custom form systems with dynamic regulatory logic and clinical pathways using state machines and logic engines.
- Integrated AI API integration and retrieval-based workflows for document analysis, clinical coding suggestions, and clinical decision-support features.
- Designed and implemented backend APIs, custom servers, and secure data pipelines to support document ingestion, processing, and clinical workflow automation.
- Built WebSocket-driven document analysis and generation workflows with concurrency-safe state handling to ensure reliable processing and support review or resubmission after completion.

Software Engineering Consultant *July 2016 – December 2021*

SRC Inc. (Various Client Projects) *Southern California*

- Delivered full-stack solutions for e-commerce, sports betting, and retail technology clients; specialized in high-performance Node.js backends and React frontends.
- Upgraded enterprise e-commerce platform with custom inventory management and disaster-recovery protocols, reducing downtime and maintenance by 95% from their old legacy system.
- and 3D interactive prototypes supporting 50K+ concurrent users.
- Built enterprise retail integrations with dynamic pricing engines, catalog management, and bulk import pipelines processing 100K+ SKUs.
- Improved CI/CD reliability across multiple client platforms using Docker and Azure, reducing deployment failures by 60%.

Senior Software Engineer (Contract) *July 2024 – December 2025*

Juniper Health Solutions · Remindfy AI · Monster Reviews *Remote*

- Built trauma registry validation and submission workflows for multi-state healthcare reporting systems.
- Designed and exposed back-end endpoints for client-side access to validation, document generation, and registry submission features.
- Built submission and validation pipelines for registry data, improving reliability, usability, and accuracy across healthcare reporting processes.
- Improved application responsiveness through paging, filtering, and caching enhancements.
- Automated registry processing workflows to reduce manual effort and improve operational efficiency.

TECHNICAL SKILLS

Languages: F#, TypeScript, JavaScript, C#, Python, Rust, Elm, Kotlin, Swift

Frontend: React, Elmish, Bulma, Bootstrap, Tailwind CSS, DaisyUI, HTML/CSS

Backend: Node.js, .NET, Azure Functions, Express, REST APIs, GraphQL

Cloud & DevOps: Azure, Digital Ocean, Docker, Kubernetes, Terraform, Helm, CI/CD, GitHub Actions

Databases: PostgreSQL, SQL Server, MongoDB, Redis, Vector DBs

AI / ML: OpenAI API, LangChain, RAG Systems, Fine-tuning, Prompt Engineering

EDUCATION

Associate of Applied Science, Computer Science *2014*

SUNY Orange